Contoso University of Engineering (CUE), which is Pakistan's premier technical university, is planning to move their datacenter to Microsoft's Azure Cloud, which will allow them to innovate faster and to reduce costs.

CUE's student registration solution is comprised of multiple on-premises applications which are currently deployed on-premises:

1. Linux Server farm with six servers behind a load balancer, hosting various java and .net web apps.
    a. One of the .net core apps hosts the university's main website. This website is also used to serve department catalogues in PDF formats.
    b. This app is also used to gather prospective student information which is then used by the university to send them relevant information. All information is stored in the backend MySQL Database.
2. Windows Server farm hosting student registration app built on .Net Framework 4.8 with data hosted on Postgres SQL Server. Currently during registration and various other peak times, students experience extreme delays and timeouts.
    a. CUE's IT staff has pointed out issues in application servers as well as database servers which are unable to scale properly to handle the required load.
3. Another load balanced farm with middleware components based on .net core, which are invoked by the applications to perform backend functionality such as emails, running jobs, integrating with external systems.
4. CUE's application infrastructure contains SQL server supporting some of the apps hosted on Linux and windows. It has multiple jobs which are scheduled to run at various intervals during the day.
5. There is also a test environment where test version of all applications is deployed along with their databases.
6. All student credentials are stored on CUE's On-Premises ADDS.

CUE's various departments' catalogues are downloaded by prospective students from all over the world, but primarily from North American and Europe. Due to the size of the catalogues, users have reported extremely slow download speeds.

Another key requirement by the CUE security team is to segregate all databases to their separate virtual network to provide better isolation and security controls whenever required.

Tasks:

1. Draw the current/as-is architecture of the CUE's application platform.
2. You need to migrate the all the applications and databases to the cloud.
3. You need to minimize and reuse compute wherever possible.
4. You need to utilize Azure Native PaaS services wherever possible, and reuse them, if possible.
5. Provide a cloud native solution for testing of new application versions and easy rollout to production.
6. Provide solution for the slow catalogue downloads for the worldwide users.
7. Provide detailed solution diagram for all components, as well as brief explanation for all design decisions.
8. Provide the identity solution which will allow students, as well as outside IT vendors working on the application to be able to login to the application from anywhere.